

Attributes	Description	Logic
ScheduleID	Unique Id	Random number
CarrierCode	Carrier code	Taken from Config data
FlightNumber	Flight number	Taken from City Pair data
AircraftType	Aircraft type	Taken from Aircraft data
AircraftTailNumber	Aircraft Register number	Taken from Aircraft data
DepartureAirport	Departure City	Taken from City Pair data
ArrivalAirport	Arrival City	Taken from City Pair data
DepartureTime	Departure time	random time
ArrivalTime	Arrival time	random time
StartDate	Effective date	random date
EndDate	Discontinue date	random date
Status	Schedule Status	Taken from Aircraft data
Sunday	Weekday	randbetween [0,1]
Monday	Weekday	randbetween [0,1]
Tuesday	Weekday	randbetween [0,1]
Wednesday	Weekday	randbetween [0,1]
Thursday	Weekday	randbetween [0,1]
Friday	Weekday	randbetween [0,1]
Saturday	Weekday	randbetween [0,1]
Frequency_per_week	count of weekday	sum of weekdays
Frequency	Frequency pattern	FrequencyPattern {1234567}
NoOfDepartures	count of departure dates	count of departure dates
Departure Dates	Departure dates in lists	List of dates between the start date and end date based on selected weekday

Attributes	Description	Logic
InventoryId	Unique ID	Random unique number
ScheduleId	Flight Scheudule ID	Taken from schedule data
AircraftType	Aircraft Model	Taken from schedule data
FlightNumber	Flight number	Taken from schedule data
DepartureDate	Journey date	Taken from schedule data
TotalCapacity	Actual Capaacity of Aircraft	Taken from config data
TotalInventory	Total Inventory	10% to 20% variation of Total capacity
BookedInventory	Booked Inventory	$\text{duration} = \text{departureDate} - \text{currentDate}$ $\text{load_factor} = \text{randombetween}(0.85, 0.1)$ $\text{booked_inventory} = (365 - \text{duration} / 365) * \text{totalInventory} * \text{load_factor}$
Oversold	Oversold	$\text{booked_inventory} - \text{total_capacity}$
AvailableInventory	Available Inventory	$\text{total_inventory} - \text{booked_inventory}$
FirstClass	Capacity of First Class	10% of total_capacity
BusinessClass	Capacity of Business Class	20% of total_capacity
PremiumEconomyClass	Capacity of Premium Economy	30% of total_capacity
EconomyClass	Capacity of Economy class	40% of total_capacity
FC_TotalInventory	Total inventory of First class	10% - 20% variation from first class total capacity
FC_BookedInventory	Booked Inventory of first class	same formula for booked inventory
FC_Oversold	oversold seats for first class	$\text{fc_booked_inventory} - \text{fc_total_capacity}$
FC_AvailableInventory	Available Inventory of firstclass	$\text{fc_total_inventory} - \text{fc_booked_inventory}$
BC_TotalInventory	Total inventory of business class	10% - 20% variation from business class total capacity
BC_BookedInventory	Booked Inventory of first class	same formula for booked inventory
BC_Oversold	oversold seats for business class	$\text{bc_booked_inventory} - \text{bc_total_capacity}$
BC_AvailableInventory	Available Inventory of business class	$\text{bc_total_inventory} - \text{bc_booked_inventory}$
PC_TotalInventory	Total inventory of premium economy class	10% - 20% variation from premium economy class total capacity
PC_BookedInventory	Booked Inventory of premium economy class	same formula for booked inventory
PC_Oversold	oversold seats for premium economy class	$\text{pc_booked_inventory} - \text{pc_total_capacity}$
PC_AvailableInventory	Available Inventory of premium economy class	$\text{pc_total_inventory} - \text{pc_booked_inventory}$
EC_TotalInventory	Total inventory of economy class	10% - 20% variation from economy class total capacity
EC_BookedInventory	Booked Inventory of economy class	same formula for booked inventory
EC_Oversold	oversold seats for economy class	$\text{ec_booked_inventory} - \text{ec_total_capacity}$
EC_AvailableInventory	Available Inventory of economy class	$\text{ec_total_inventory} - \text{ec_booked_inventory}$
FC_CD	Class code and distributed seats of first class	class_code and percentage of distribution taken from config file
BC_CD	Class code and distributed seats of business class	class_code and percentage of distribution taken from config file
PC_CD	Class code and distributed seats of premium economy class	class_code and percentage of distribution taken from config file
EC_CD	Class code and distributed seats of economy class	class_code and percentage of distribution taken from config file